MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)

MG1DSCSTA100 - Fundamentals of Statistics and Data Visualisation

Duration: $1\frac{1}{2}$ Hours Maximum: 50 Marks

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions

*Answer any **seven** questions. Each question carries **2 marks.***

1. Define pie diagram. [Remember][CO1]
2. What is meant by sampling error? [Understand][CO2]
3. What is the median of 4, 2, 7, 3, 10, 9, 13? [Understand][CO2]
4. Distinguish between absolute and relative measures of dispersion. [Understand][CO2]
5. Define mean deviation. Mean deviation is calculated by considering ------ deviations. [Understand][CO2]
6. If both standard deviation and coefficient of variation are 15, find the mean. [Understand][CO2, CO7]
7. If the first three raw moments about 5 are 2, 20, 40, then find the first 3 central moments. [Understand][CO1]
8. Give the expression of Pearson's measure of skewness. [Understand][CO2]
9. Mention any two advantages of rank correlation. [Understand][CO3]
10. State the classical definition of probability. [Understand][CO5]

(2x7=14)

Part B

Short Essay questions

*Answer any **four** questions. Each question carries **6 marks**.*

11. Evaluate the difference between nominal and ordinal scale data giving examples. [Understand][CO1]
12. Explain lottery method and random number table method in simple random sampling. [Understand][CO2]
13. Define arithmetic mean. The following are the marks of 10 students in an examination: 20, 40, 65, 83, 34, 51, 70, 62, 88, 94. What is the average mark of the students? [Understand][CO2]
14. If 3x+2y-80=0 is the regression equation of X on Y and 2x+3y-70=0 is the regression equation of Y on X, then find the two regression coefficients, and hence the correlation coefficient. [Apply][CO3, CO4, CO7]
15. (i) Define mutually exclusive and exhaustive events with examples. (ii) Two dice are rolled. Identify the events of getting a sum of 7 and same number. Are these events mutually exclusive? Justify your answer. [Understand][CO5]
16. Let A and B be two events associated with an experiment and suppose P(A)=0.5 while P(A or B)=0.8. Let P(B)=p. For what values of *p* are (i) A and B mutually exclusive (ii) A and B are independent. [Apply][CO5, CO6, CO7]

(6x4=24)

Part C

Essay questions

*Answer any **one** question. Each question carries **12 marks**.*

17. Define primary data. Discuss different methods of primary data collection. [Understand][CO1]
18. (i) Define correlation and briefly discuss its different types. (ii) Calculate the Karl Pearson's correlation coefficient between X and Y:

X	5	8	12	18	7
Y	12	10	22	17	14

[Analyse][CO3]

(12x1=12)